

STRINGS – INPUT & BASICS

```
cin >> s; // single word
s.length(); // or s.size()
getline(cin, s); // full line w/ spaces
s.empty(); // true if s==""
```

TRAVERSE

```
for (int i=0;i<s.size();i++) cout<<s[i]; for (char c : s) cout<<c;
```

CASE CONVERSION

```
s[i] = toupper(s[i]); s[i] = tolower(s[i]);
transform(s.begin(), s.end(), s.begin(), ::toupper); // whole string upper
transform(s.begin(), s.end(), s.begin(), ::tolower); // whole string lower
```

SUBSTRING & SEARCH

```
s.substr(pos, len); // from pos, len chars
s.substr(pos); // from pos to end
s.find("abc"); // index or string::npos
if (s.find('c') != string::npos) // found check
```

COMPARE & CONCATENATE

```
s1 == s2; // equality s1 < s2; // lexicographic
string r = s1 + s2; // concat s += "abc"; // append
```

INSERT / ERASE

```
s.insert(pos, "abc"); s.erase(pos, len); // erase len chars from
pos
```

CHARACTER CHECKS

```
isalpha(c); isdigit(c); isupper(c); islower(c); isalnum(c); isspace(c);
```

CHAR / NUMBER CONVERSION

```
int d = s[i] - '0'; // char to digit char c = '0' + d; // digit to char
int x = stoi(s); // string to int string s = to_string(x); // number to
string
long long x = stoll(s); // string to long long
```

FREQUENCY COUNT

```
int freq[26] = {0}; for (char c : s) freq[c-'a']++; // letter frequency
```

PALINDROME CHECK

```
string t = s; reverse(t.begin(), t.end()); bool isPalin = (s == t); // reverse
method
for (int i=0,j=s.size()-1; i<j; i++,j--) if (s[i]!=s[j]) { /*not palin*/ } // two-
pointer method
```

ANAGRAM CHECK

```
string a=s1,b=s2; sort(a.begin(),a.end()); sort(b.begin(),b.end()); bool
isAnagram=(a==b);
```

SPLIT BY DELIMITER

```
vector<string> tokens; stringstream ss(s); string tok;
while (getline(ss, tok, ' ')) tokens.push_back(tok); // split on space
```

SORT CHARACTERS OF A STRING

```
sort(s.begin(), s.end());
```

BUILD REPEATED-CHAR STRING & TRIM SPACES

```
string s(n, 'a'); // string of n 'a' characters
s.erase(remove(s.begin(), s.end(), ' '), s.end()); // remove all spaces
```